where, in our case, the following applies to each component α_i:

$$\alpha_{\mathrm{i}} \sim \omega_{\mathrm{i}}. \tag{2.29}$$

The respective weights are determined with the help of an additional prior distribution:

$$\mathbb{P}(\boldsymbol{\phi}) = \mathbb{P}(\boldsymbol{\omega}) = \mathrm{Dir}(\omega_1, ..., \omega_{\mathrm{K}} \mid \frac{1}{2}, ..., \frac{1}{2}). \tag{2.30}$$

Since we have determined a priori equally distributed weights, this relationship simplifies. However, if certain taxa are given a higher or lower weight at the beginning, this must be taken into account accordingly at this point.

Fig. 2.17 summarizes graphically how our local reconstruction framework works in general. The boxes in the first row contain the input variables, while the variables in the white boxes are not inferred during the MCMC simulation. For example, the parameters of the transfer functions are defined in Sect. 2.5. The middle upper gray box describing the inference of the taxa weights can be optionally determined via Gibbs or rwMH sampling. This is done by comparing the sampled climate reconstructions with other climate data and/or reference curves and constraining them accordingly. These comparisons are made using the Metropolis-Hastings sampling, represented by the lower dashed area.

2.5 Transfer functions via machine learning

This section describes the development of transfer functions. The main attention is paid to the probability distribution $\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi})$ from Eq. 2.15. We start with the idea of (Kühl et al., 2002), who uses multivariate Gaussian distributions for the presence of plants. For this purpose, analogous to this work, they took plant distribution maps as for the European olive tree in Fig. 2.18 (c) and assigned certain climate variables to them. In our case, these are the distributions of $\mathrm{P_{ANN}}$ and $\mathrm{T_{ANN}}$ over the years 1961 – 1990 AD from the CRU 4.01 dataset (Harris and Jones, 2017) and are shown in Fig 2.18 (a) and (b). These data, along with those from Fig. 2.18 (c), are presented in the climate feature space in Fig. 2.19. Sect. 2.1.1 already compiles the relationship between climate variability such as Heinrich, Bond, and DO events and isotopic information from speleothems. These fluctuations occurred within several decades to centuries and affected temperature and precipitation in regions of Eurasia and North America (Asmerom et al., 2010; Wagner et al., 2010; Genty, 2003; Benson et al., 1996). The assumed relationship between speleothems and these two climate variables differs from study to study. In the Levant, these are often used as proxies for the mean annual precipitation (Bar-Matthews et al., 2003; Cheng et al., 2015; Sánchez-Fernández et al., 2018; Bradley, 2014). However, in Central and Northern Europe it alternates between $\mathrm{T_{ANN}}$ and $\mathrm{P_{ANN}}$ (Mauri et al., 2015; Lauritzen and Lundberg, 1999). Since plants are also sensitive to such climatic variations (Gams,

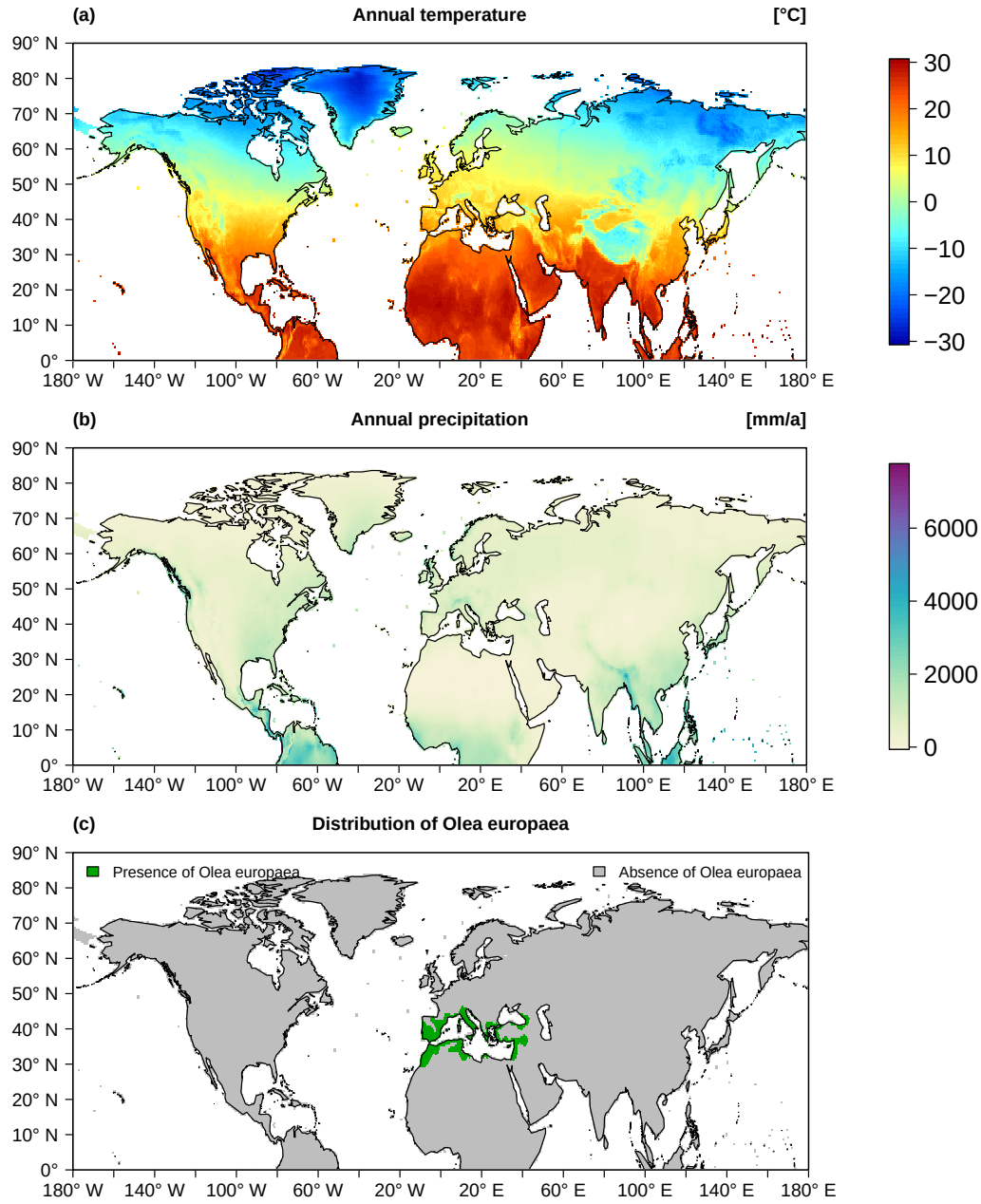


Figure 2.18: Maps of the Northern Hemisphere with annual precipitation (a), annual temperature (b), and the geographical distribution of Olea europaea (c). The climate data are based on those from CRU 4.01 and the plant distribution is according to Meusel et al., 1974.

1964), a relationship between plant distributions, temperature and precipitation variables is assumed in this work. This is done in some studies based on the PDF method (Kühl et al., 2002; Kühl and Litt, 2003; Gebhardt et al., 2008). Thus, the climate variables considered are intended to represent a compromise between the various proxies.

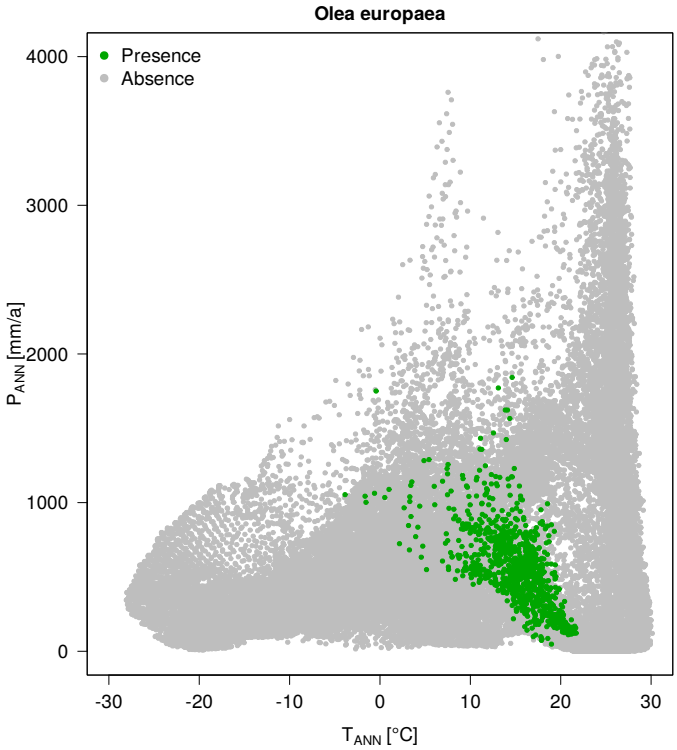


Figure 2.19: Summary of the data points from Fig. 2.18 (a) - (c) into a climate space that represents the feature space of the machine learning algorithms.

After Kühl et al., 2002 successfully used multivariate Gaussian distributions as transfer functions, Gebhardt et al., 2004 compared normal distributions, kernel densities, and Gaussian mixture models. Daniel Simonis, 2010 was the first to use precipitation in addition to temperature. Using the inverse cumulative distribution function (CDF) method (Gentle, 2004), these can be transformed into a standardized normal distribution. Standardization and scaling of input variables is generally an important concept in the computations of machine learning models (Butcher and Smith, 2020). Therefore, we also use the inverse CDF method before passing the climate data to the ML algorithms. To calculate the best number of components of the mixture model, Daniel Simonis, 2010 uses the Bayesian information criterion (BIC), in contrast to Gebhardt et al., 2004, who take the log-likelihood. Also in Chevalier et al., 2014, PDFs are applied as transfer functions that consider only the presence information in the climate space (feature space). Stolzenberger, 2011, Stolzenberger, 2017, Thoma, 2017, and Weitzel et al., 2019 instead used logistic regression that accounts for not only the presence but also the absence of plants. In Litt et al., 2012, Stolzenberger, 2017 and in Thoma, 2017, Quadratic Discriminant Analysis (QDA) is applied with respect to three biomes. The absence of these biomes (or unknown biomes) has not been used as an additional feature in previous work.

The objective of this work is to systematically test a variety of possible methods and select the most

appropriate algorithm for the task at hand. For this purpose we use the R package caret (Jed Wing et al., 2019). This stands for **c**lassification **a**nd **re**gression **t**echniques and provides a variety of models that can be used to solve corresponding problems. In our case, it is a binary (presence or absence of a taxon) classification problem. The package provides a simple way to compare the selected models via cross-validation. In this process, the provided data (cf. Fig. 2.18) are split into a training and a validation dataset. Cross-validation is performed on the training data (James et al., 2013), which in our case accounts for 70 % of all data. Accordingly, this training data is split into a certain number (k-folds) of training and validation datasets. This partitioning process can be repeated as often as desired and leads to statistical verification distributions from which the performance of the models is derived. Cross-validation is also performed for a certain number of different parameters for the respective machine learning algorithms (model tuning). The entire process is very easily accessible in caret and runs completely automatically after the initial parameters have been defined. The remaining 30 % (hold-out set) are used to validate the models obtained by cross-validation on the remaining 70 %. This has the advantage that they can be tested on an independent data set, further minimizing the risk of overfitting. More precisely, the winner is determined from the first half and its final score from the second half of the hold-out set. Thus, it is quite possible that the final selected transfer function differs from the winner of the cross-validation.

As shown in Fig. 2.18 (c), the groups for presence (minority class) and absence (majority class) are unbalanced. One could reduce the size of the entire map section so that the groups are balanced. However, the models then deliver significantly worse and sometimes more unrealistic results. This problem is discussed for example in Thoma, 2017 or in Weitzel et al., 2019. Thus, a model could provide higher probabilities of occurrence on the one hand where the plant does not occur in the feature space, and on the other hand where the climate values are biologically unrealistic. When the map section is enlarged, this problem recedes, especially if the absence values can serve as a boundary. This is the case when the occurrence domain is enclosed by the absence domain in the feature space. The reduction of the map section is analogous to the techniques of random under-sampling (Hoens and Chawla, 2013), wherein the majority class is randomly reduced to the size of the minority class, potentially losing important information. In contrast, random oversampling of the minority class risks overfitting. To solve this problem, the **S**ynthetic **M**inority **O**versampling **Te**chnique (SMOTE) is used (Bowyer et al., 2011). Here, a minority class instance a is first randomly selected and its k-nearest minority class neighbors are determined. A line segment is then formed between one randomly selected k-nearest neighbor b in feature space. Here, a synthetic instance of the minority class is created by selecting a random point along this line (Hoens and Chawla, 2013). SMOTE can only do this with one minority class at a time. The new synthetic data thus generated are the input data for the calculation of the transfer function in the ML competition. It should be noted that only the training data is processed with SMOTE. For the model verification on the hold-out set, the original data are used. The following is a brief description of the ML algorithms used in this work.

Generalized Linear Model (GLM)

Logistic regression, listed under the GLM, is similar to linear regression (Agresti, 2003). However, here the outcome variable is categorical and not continuous (Fahrmeir and Tutz, 1994). In logistic regression, it is assumed that the outcome variable follows a binomial distribution and the logistic probability (logit) is accordingly described by a linear function of the logistic regression coefficients $\boldsymbol{\psi}$. The inverse function of the logit is also called the logistic function and is:

$$\mathbb{P}(\boldsymbol{P_s} \mid \boldsymbol{C}, \boldsymbol{\psi}) = \frac{\exp(\boldsymbol{C\psi})}{1 + \exp(\boldsymbol{C\psi})}. \tag{2.31}$$

Regression coefficients are obtained using maximum likelihood estimation for each taxon (Agresti, 2003). The values of the GLM can be interpreted as probabilities for the presence or absence of the taxa. In this thesis, quadratic logistic regression is used analogously to Stolzenberger, 2017, Thoma, 2017 and Weitzel et al., 2019. The resulting bilinear terms are crucial for additional flexibility in the feature space and can be written as follows for the first selected taxon $P_{1,s}$:

$$\mathbb{P}(P_{1,s} \mid \boldsymbol{C}, \boldsymbol{\psi}) = \text{logit}(\psi_1^T + \psi_2^T C_1 + \psi_3^T C_2 + \psi_4^T C_1 C_2 + \psi_5^T C_1^2 + \psi_6^T C_2^2). \tag{2.32}$$

When these quadratic terms are not used, the GLM performs significantly worse than the other ML algorithms (not shown), even though they do not use the additional information. They are deprived of these, as this hardly leads to any improvement (not shown). The GLM therefore always has an advantage, but should still be compared in order to be able to establish a reference to the earlier studies.

Artificial Neural Network (ANN)

In the broadest sense, the idea of Artificial Neural Network dates back to the 18th century (Schmidhuber, 2015). The structures described there are variants of logistic regression. Such an ANN is also called a "logistic perceptron" (Spackman, 1991). The input values (here: climate variables) are transferred to the output via a simple interface. Analogous to Eq. 2.31, the input is provided with weights (regression coefficients) and processed via a so-called activation function (here: logistic function). The counterpart to the coefficients of the axis intercepts is called bias in ANNs. The further development of ANNs began in the middle of the 20th century (Mcculloch and Pitts, 1943). The architecture of these networks was modeled on the biological nervous system. The different layers consist of artificial neurons (interfaces) that "fire" at certain values using the activation functions and thus pass on their signal. In addition to the input and output layer, a so-called hidden layer is also used. This allows, in contrast to the logistic perceptron, the modeling of more complex relationships between input and output (Ayer et al., 2010). Over the course of time, the network structure has been continuously developed (Schmidhuber, 2015). Among the best known networks is the recurrent neural network (Rumelhart et al., 1986). In contrast to the feedforward neural network (FNN), the

connections of neurons of one layer to neurons of the same or a previous layer are guaranteed. This opened up the possibility that ANNs could also be used for speech and character recognition. In image recognition the so-called Convolutional Neuronal Networks are used (LeCun et al., 1989). For a detailed description of the history of ANNs, the reader is referred to the work of Schmidhuber, 2015.

Comparatively simple classification problems arise in this work, so relatively simple network structures can be used. These give similarly good results with much less computational cost, and the risk of overfitting is generally lower with simpler structures. The caret package offers a wide range of different ANNs, including those based on Python. After initial tests, the ANN from the R package nnet is chosen in this thesis. It is an FNN that allows one hidden layer with an arbitrary number of hidden neurons (Venables and Ripley, 2002). This enables much more complex correlations to be found than with the GLM described above. Moreover, for our classification problem, it gives partially better results than the multilayer perceptrons provided by the caret package and the training time is also lower.

Discriminant analysis

Discriminant analysis involves the development of discriminants, i.e. linear combinations of independent variables that discriminate the categories of the dependent variable (James et al., 2013). Linear Discriminant Analysis (LDA), for example, first extracts linear constructs (discriminants) that maximize separation between groups and then uses them to perform a Gaussian classification. This step can also be used as data reduction. At baseline, LDA assumes normality as well as variance-covariance homogeneity of the groups. Then Bayes' theorem (see Eq. 2.7) is used to assign the observations to groups based on the discriminants. In our case, a corresponding equation could look like this:

$$\mathbb{P}(\boldsymbol{P_{1,s}} \mid \boldsymbol{C}, \boldsymbol{\psi}) = \frac{\mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_1}, \boldsymbol{\Sigma}) \cdot \mathbb{P}(\boldsymbol{P_{1,s}})}{\sum_{i=1}^{2} \mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_i}, \boldsymbol{\Sigma}) \cdot \mathbb{P}(\boldsymbol{P_{i,s}})}.$$

$\boldsymbol{\mu_1}$ is the estimated mean value of group 1, which is assumed to be multivariate normally distributed, e.g. the occurrence of plants. Group 2 is interpreted accordingly as absence. $\boldsymbol{\Sigma}$ is the joint covariance matrix and $\boldsymbol{P_{i,s}}$ the respective population group proportions (Johnson and Wichern, 2007) of the groups. The Bayesian approach, based on discriminants, allows the use of separate group discriminant matrices in addition to the standard use of a covariance matrix. This brings us to Quadratic Discriminant Analysis (QDA). QDA is a modification of LDA that accounts for heterogeneity in the covariance matrices of the groups. Therefore, the covariance matrices $\boldsymbol{\Sigma_i}$ estimated separately for each group are used here:

$$\mathbb{P}(\boldsymbol{P_{1,s}} \mid \boldsymbol{C}, \boldsymbol{\psi}) = \frac{\mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_1}, \boldsymbol{\Sigma_1}) \cdot \mathbb{P}(\boldsymbol{P_{1,s}})}{\sum_{i=1}^{2} \mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_i}, \boldsymbol{\Sigma_i}) \cdot \mathbb{P}(\boldsymbol{P_{i,s}})}.$$

In this thesis, QDA provides much better results compared to LDA due to the higher flexibility of the respective plant distributions (not shown). The latter is therefore removed from the competition. Next,

we consider Mixture Discriminant Analysis (MDA). MDA can be seen as an extension of LDA and thus of the work of Kühl et al., 2002. It modifies the within-group multivariate density of predictors by a mixture (i.e., a weighted sum) of multivariate normal distributions (Rausch and Kelley, 2009):

$$\mathbb{P}(\boldsymbol{P_{1,s}} \mid \boldsymbol{C}, \boldsymbol{\psi}) = \frac{\sum_{j=1}^{G_1} \gamma_{j,1} \mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_{j,1}}, \boldsymbol{\Sigma_{j,1}}) \cdot \mathbb{P}(\boldsymbol{P_{1,s}})}{\sum_{i=1}^{2} \sum_{j=1}^{G_i} \gamma_{j,i} \mathcal{N}(\boldsymbol{C} \mid \boldsymbol{\mu_{j,i}}, \boldsymbol{\Sigma_{j,i}}) \cdot \mathbb{P}(\boldsymbol{P_{i,s}})}.$$

G represents the clusters and $\gamma_{j,i}$ the mixture weights within each group. In principle, this approach brings two further advantages over LDA and QDA: (1) more accurate classification by modeling multivariate nonnormality or nonlinear relationships between variables within each group, and (2) the possible detection of subgroups in each group (Rausch and Kelley, 2009).

Decision tree-based algorithms

Algorithms based on decision trees are often used for classification of ecological data (De'ath, 2007). They are divided into so-called branches and leaves, which subdivide the feature space by yes-no queries. The disadvantage of simple decision trees is their inaccuracy. Often their prediction accuracy is significantly worse than the achievable (Hastie et al., 2009). This can be increased dramatically in some cases if bagged or boosted trees are used. The random forest, for example, uses bagging based on decision trees (Breiman, 2001). This typically involves determining several hundred bootstrap datasets from the feature space. If multiple features are available, subsets are drawn randomly from these datasets and the decision trees are finally calculated on the basis of these subsets. This abundance of different trees forms the random forest. Bootstrapping the data and using the aggregate to make a decision is called bagging. Now, when a prediction is made with a new data point, the votes of all decision trees are collected together, from which a probability for that data point can be determined. However, in the present case, the random forest is not the most appropriate algorithm because the dataset consists of the two features T_{ANN} and P_{ANN}. If more features were available, it would make sense to include the random forest in such a machine learning competition.

Boosted trees are similar to random forests in that a number of decision trees are computed and considered. However, only shallow trees are considered here. For example, in Adaptive Boosting (Freund and Schapire, 1996), so-called stumps (trees with two leaves) are almost always used as "weak" learners (Friedman et al., 2000) and in the Gradient Boosting Machine (GBM) (Friedman, 2001), the "depth" of the trees is also specified at the beginning. To reduce bias and variance, each decision tree is determined as a function of its predecessor and weighted according to its accuracy. This combining of many model results can decrease bias and, to some extent, variance. However, boosting also loses some advantages over simple decision trees, such as speed, interpretability, and, in the case of Adaptive Boosting, robustness to overlapping group distributions and, most importantly, mislabeling of the training data. A GBM is a generalization of tree-boosting that attempts to mitigate

these problems to create an accurate and effective standard procedure (Hastie et al., 2009). For this reason, GBM is chosen in this thesis to introduce a machine learning algorithm based on decision trees.

Support Vector Machines (SVMs)

Support Vector Machines determine a function that divides the input data into classes (Jergensen et al., 2020). In our case, the support vectors are the correctly classified data points closest to the separation line between the two classes. The distance between this line and the support vectors is called margin, which is maximized. Although these algorithms provide the best results in some cases, they consistently failed in comparison with the other algorithms because their computation time is disproportionately long. Similar difficulties with SVMs are also found in Jergensen et al., 2020, where a ML competition for forecast models of convective storms is presented. For this reason, we exclude SVMs from the competition.

2.6 Spatial climate reconstructions

Finally, in this work we aim to create climate field reconstructions (CFRs) based on our local climate reconstructions. Let us first briefly review the history of spatial reconstructions in the paleoclimatological context.

For reconstructions within the last two millennia, there are a number of methods based on regression techniques (Mann et al., 1998). Other examples include Cook et al., 1999 with point-to-point regression, Ammann and Wahl, 2007 with principal component regression, Arrigo et al., 2006 with univariate linear regression, Hegerl and Zwiers, 2011 with total least squares, and Mann et al., 2008 with regularized expectation maximization. A different approach is used by Gebhardt et al., 2008. They combine pollen and macrofossil samples with an advection-diffusion model, which is driven by insolation changes between the present and past time slices. A corresponding cost function is minimized and thus the climate of the early Eemian (ca. 125 cal ka BP) is reconstructed. Simonis et al., 2012 built on this idea and extended the model by importance sampling and reconstructed the periods 6 cal ka, 8 cal ka, 12 cal ka, and 13 cal ka. Davis et al., 2003 and Mauri et al., 2015, in turn, calculate CFRs for different time periods using pollen samples primarily from EPD. To spatiotemporally interpolate these, they use 4D thin plate splines.

Stolzenberger, 2017 and Weitzel et al., 2019 combine the pollen and macrofossil dataset of Simonis et al., 2012 with a MH climate simulation ensemble from the Paleoclimate Modelling Intercomparison Project 3 (PMIP3; Braconnot et al., 2011). Each individual model setup is considered a separate ensemble member. The entire procedure aims to constrain the spatial structures of physically possible climate states for a given external forcing. In particular, Weitzel et al., 2019 systematically investigate this procedure using a Bayesian framework. In doing so, they present several ways in which spatial